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CCBS 4024
Investments and Portfolio Management



Topic 1:
Risk and Return of a single asset

Lecturer

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(Please don’t use SOUL message to contact me.)



Housekeeping stuffs

Class disciplines:

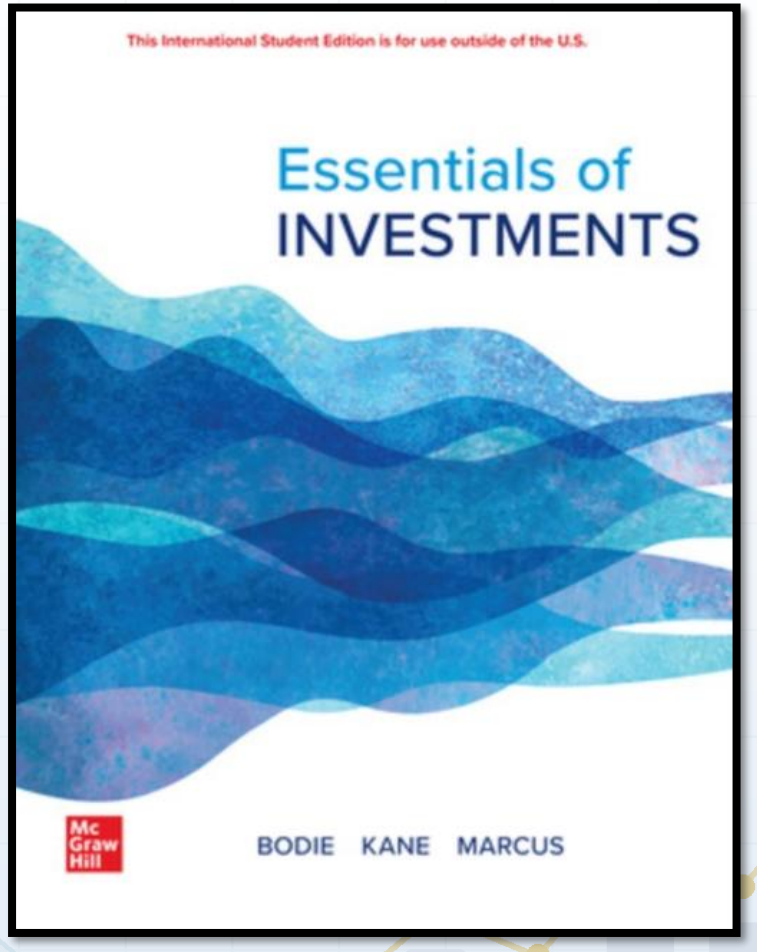
- Be punctual.
- Switch your mobile to vibrating/silent mode.
- Don’t talk loudly with your friends during lectures.
- No mobile games.
- Plagiarism (抄襲) is strictly prohibited.
- It is your responsibility to attend lecture or to catch up on any missed material should you be unable to attend.
- Any queries about the course, ask your teacher concerned.



Subject matters

Textbooks

Bodie, Kane & Marcus, Essentials of Investments, 13th ISE, McGraw Hill



Assessments

1) Final Examination in Dec/Jan 40%
2) Mid-Term Test in Nov 20%
3) 2 sets of assignments (15% each) 30%
4) Class Engagement Activities 10%

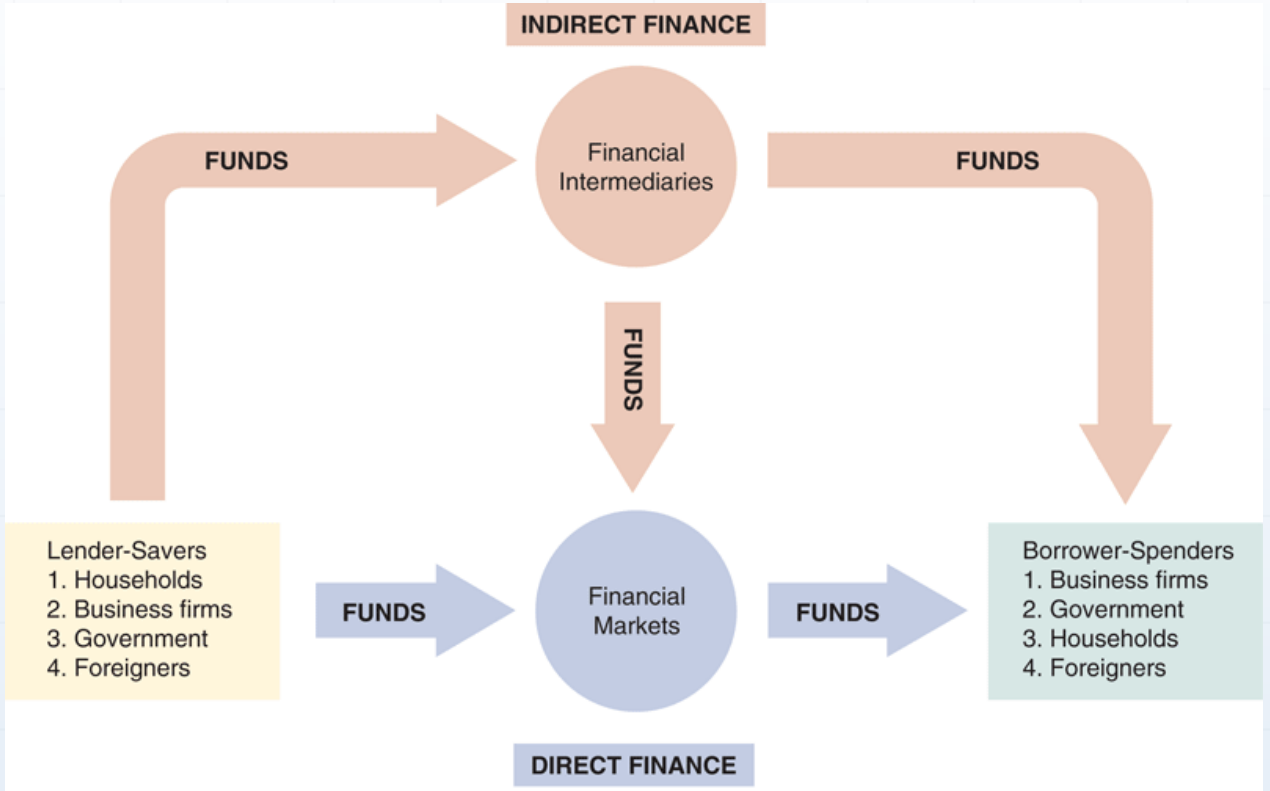
Total: 100%



Syllabus

1. Introduction to Financial Markets and the Investment Environment
2. Modern Portfolio Theory: Markowitz Model
3. Equilibrium in Capital Markets: CAPM and Arbitrage Pricing Theory (APT)
4. Fixed Income Securities and Interest Rate Risk
5. Derivative Securities
6. Trends in Investments

Financial system



The arrows show that funds flow from lender-savers to borrower-spenders via two routes: *direct finance*, in which borrowers borrow funds directly from financial markets by selling securities, and *indirect finance*, in which a financial intermediary borrows funds from lender-savers and then uses these funds to make loans to borrower-spenders.

What is Finance?

- All individuals and organizations can be classified as **surplus units (Lender-savers) or deficit units (Borrower-spenders).** Surplus units have surplus of funds which are usually households whereas deficit units are in lack of funds which are usually firms and governments.

- Finance is the processes by which money or fund is transferred (financing and investing) among businesses, individuals and governments.

- Thus, financial instruments, financial institutions and financial markets are studied in finance.



What is Investment?

Investment is about allocating money or resources with the expectation of generating a profit or return on various assets such as stocks, bonds, real estate, mutual funds, and other financial instruments.



Types of Assets

Real Assets

- Assets used to produce goods and services to consumers
- Examples: land, buildings, plant & equipment, human capital, etc.

Financial Assets

- Claims on real assets or claims to the income generated by real assets
- Examples: stocks, bonds, and derivatives

Two types of decisions in portfolio constructions

Asset allocation:

Allocate portfolio investment across broad asset classes:

1. Stocks (e.g. 50%)
2. Bonds (e.g. 30%)
3. Real estate (e.g. 15%)
4. Commodities (e.g. 5%)

Security selection:

Choosing specific securities within an asset class: e.g. HSBC

- The asset allocation decision is the primary determinant of a portfolio's return

Top-down approach vs. Bottom-up approach

- **A top-down investor makes the asset allocation decision before turning to the security decision.**
- **A bottom-up strategy chooses securities that seem attractively priced without as much concern for the resultant asset allocation.**

Separation of Ownership and Management

Ownership vs. Management

In finance, firms aim to maximize **shareholders' wealth** or share price. However, the large scale of modern corporations often requires a separation between ownership and management.

Scale Makes It Necessary

In 2022, ExxonMobil held more than **$370 billion in assets**, including $200 billion in property, plant, and equipment. With tens of thousands of shareholders, it would be impractical for individuals to take an active role in managing the firm.

The Result: Agency Problems

Owners' interests may differ from managers' interests — leading to **shirking, excessive perks, empire-building, and overly conservative strategies**.

Ways to reduce the agency problem of a firm

Several mitigating factors encourage managers to act in the shareholders' best interest:

1. **Performance-Based Compensation**
   Tying pay to share price or profit — through bonuses, stock, and options — aligns managers' rewards with owners' returns.

2. **Boards of Directors**
   Elected by shareholders, the board monitors performance and can replace managers who underdeliver.

3. **Threat of Takeovers**
   Persistent underperformance depresses the share price and invites a takeover — after which incumbent management is usually replaced.

Does ESG Benefit Shareholders?

Broader Corporate Goals

Groups of CEOs of the country's largest corporations advocate goals that address other stakeholders — employees, customers, and communities. These strategies fall under **environmental, social, and governance (ESG) investing**.

The Criticism

Some argue that when CEOs weigh ESG in their decisions, the outcome **may not maximize shareholders' wealth** or serve shareholders' interests.

The Evidence

A growing body of research shows that building ESG into the goals of the firm can **increase firm value and benefit shareholders in the long run**.

Four Core Value Drivers of ESG

How does an ESG strategy create shareholder value?

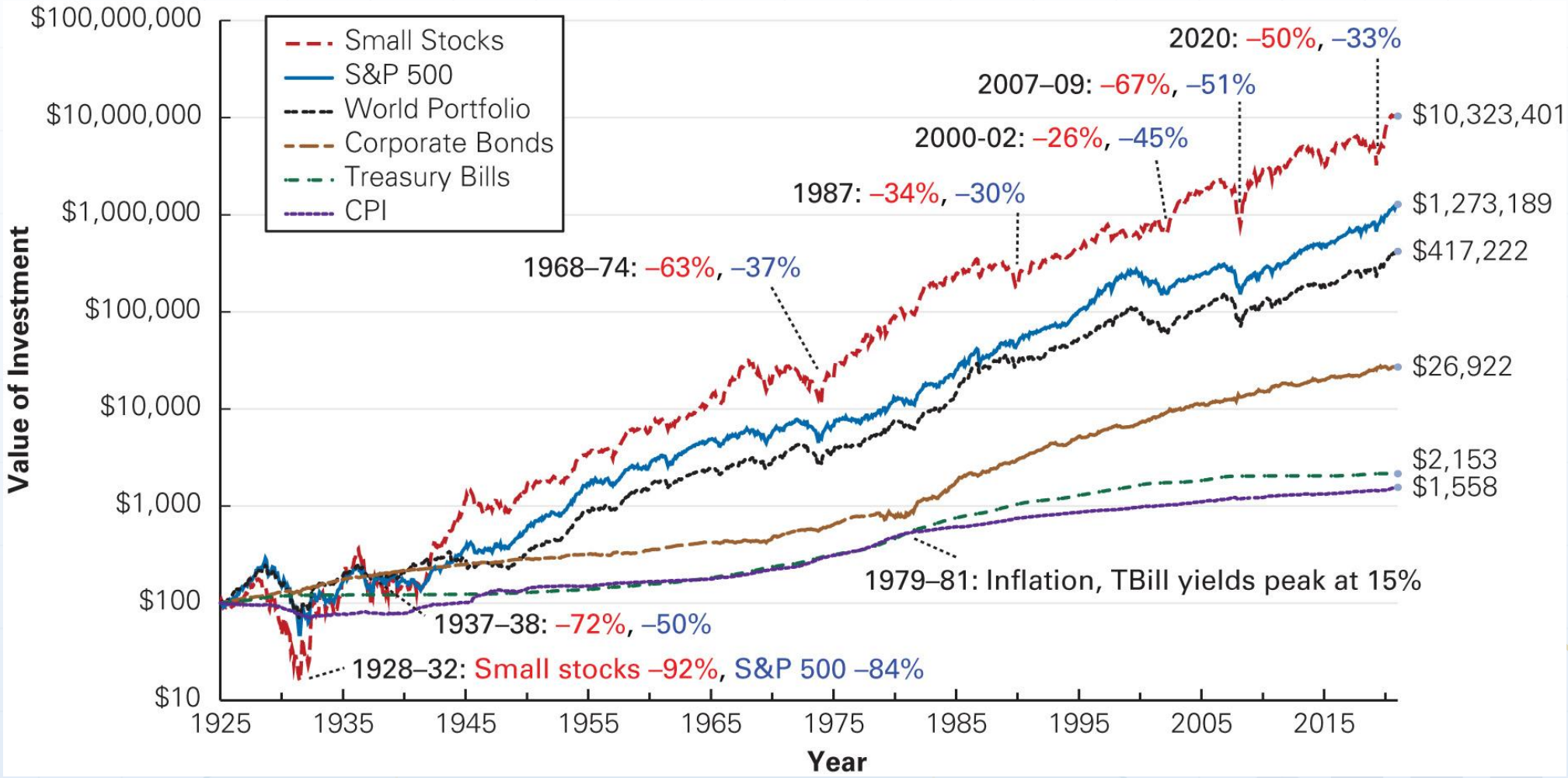
1. **Lower cost of capital:** Investors may require a lower risk premium, reducing the cost of equity and debt and increasing valuation.
2. **Operational efficiency:** Lower waste and energy use can improve margins and raise return on invested capital (ROIC).
3. **Risk mitigation:** Fewer regulatory penalties, lawsuits, stranded assets, and reputational shocks reduce downside risk in future cash flows.
4. **Talent retention and pricing power:** Better employee retention and stronger demand for sustainable products can support revenue growth.

Risk and Return: Insights from 96 Years of Investor History in the U.S.

How would $100 have grown if it were placed in one of the following investments?

1. **Standard & Poor's 500**: Leaders in their industries and among the largest firms traded on U.S. Markets
2. **Small stocks**: Securities traded on the NYSE with market capitalizations in the bottom 20%
3. **World Portfolio**: International stocks from all the world's major stock markets in North America, Europe, and Asia
4. **Corporate Bonds**: Long-term, AAA-rated U.S. corporate bonds with maturities of approximately 20 years
5. **Treasury Bills**: An investment in three-month Treasury bills

Value of $100 Invested at the End of 1925



Source: Chicago Center for Research in Security Prices, Standard and Poor's, MSCI, and Global Financial Data.

Conclusion:



_____ had the highest long-term returns, while T-Bills had the (highest / lowest) long-term returns.

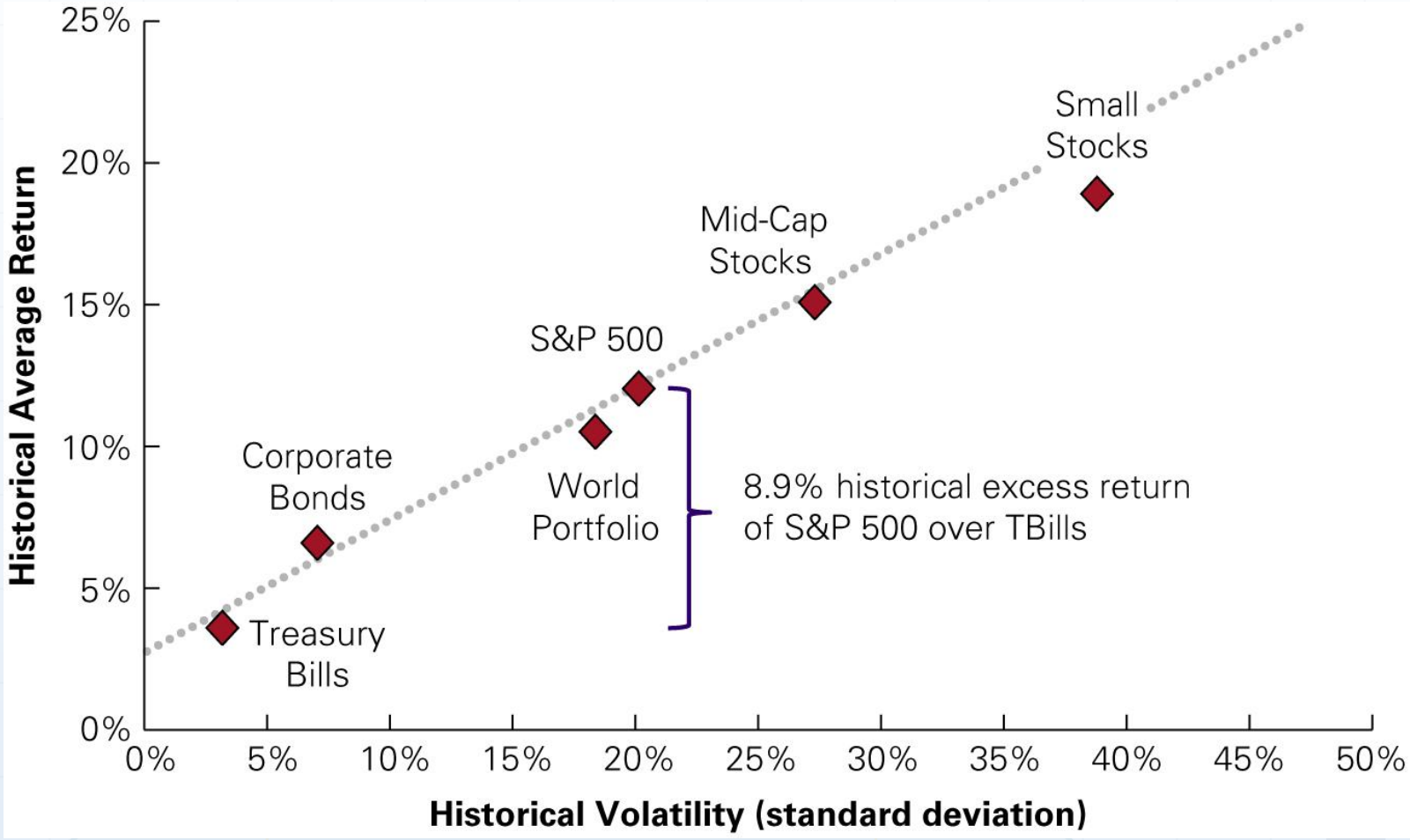


_____had the largest fluctuations in price, while T-Bills had the lowest.



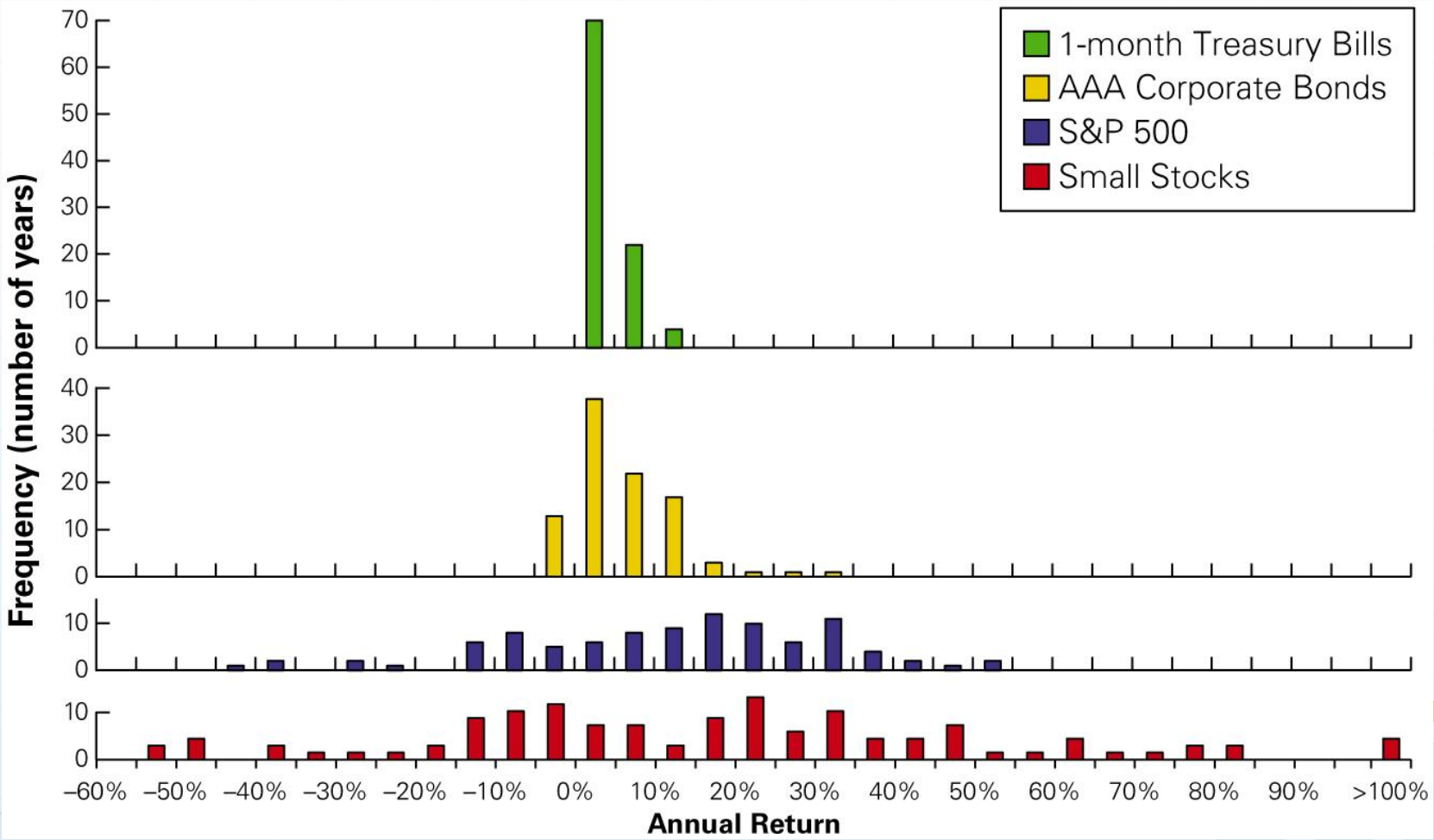
Traditional wisdom in finance: Higher _____ requires a higher _____.

The Historical Tradeoff Between Risk and Return in Large Portfolios



Source: CRSP, Morgan Stanley Capital International

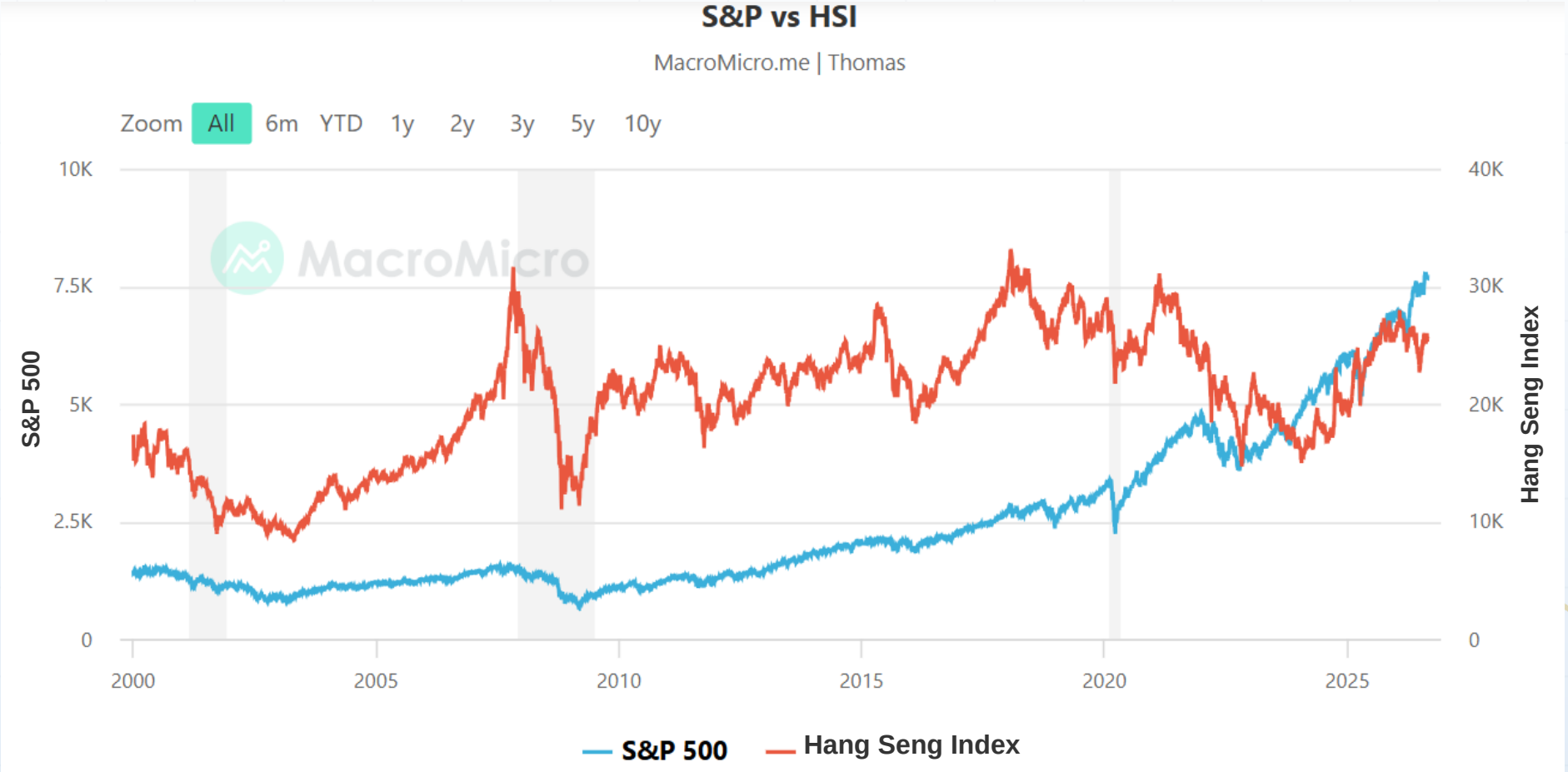
The Empirical Distribution of Annual Returns for U.S. Large Stocks (S&P 500), Small Stocks, Corporate Bonds, and Treasury Bills, 1926–2021



Importance of the stock market

- Fluctuations in stock prices can affect the economy's GDP by **affecting the spending of households and firms**.
- The stock market is an important **source of funds for corporations**. Boom stock market allow firms to raise more funds easier for making real investment.
- Stocks also make up a significant portion of household's wealth. Households spend more when their wealth increases and less when their wealth decreases.
- Stock market fluctuations can heighten **uncertainty** and lead households and firms to postpone their spendings on investments and consumptions.

Hang Seng Index vs. S&P 500 daily closing, 2000 – 2026



Source: https://en.macromicro.me/charts/24230/S-P-vs-HSI?share=41099d21a5d03bf8e3410d3fc5d75a48acaf5b8f3501eb7297eb446b9f175c57

Introduction to stock indices

- A stock index represents the average price of shares trading on a stock exchange. The average is computed using the prices of only a handful of stocks that are deemed to be the most representative of the market as a whole.

- Since the index is based on representative stocks, an upward movement in the index signals an increase in the price of a majority of stocks.

- Stock indices are also very useful in portfolio management. A broad-based stock index is normally used as a **proxy for the 'market' portfolio** and to determine the systematic risk of the portfolio. This helps portfolio managers make investment decisions.

Introduction to stock indices

- The **Dow Jones Industrial Average (DJIA)** was founded in 1896 with only 12 stocks. Over the years, more stocks were added to the index and it is now based on 30 stocks.

- In contrast, the more modern **Standard and Poor's 500 (S&P 500)** index consists of 500 stocks, while the NYSE Composite Index is based on the prices of all stocks traded on the New York Stock Exchange.

- In Hong Kong, Hang Seng Index (HSI) has _____ constituent stocks in August 2026.

Index calculation

- Stock indices are calculated using different methods for determining the average share prices. The two most widely used methods are:

  a) **price-weighting** and;
  b) **value-weighting**.



Index calculation

a) **Price-weighted indices – DJIA:**

Price-weighted indices are calculated simply as the average share price of stocks included in the index on a given day. Suppose there are two stocks in the Dow Jones Average, ABC and XYZ. Stock ABC starts at $25 a share and increases to $30. Stock XYZ starts at $100 but falls to $90. Find the initial and final value of the hypothetical DJIA.

Initial value: _____ Ending value: _____

In realty, the DJIA was calculated by adding the share prices of all the 30 largest listed stocks and then divided by 30 (called the divisor). Drawbacks of this method include:

- High-priced stocks affect the index much more than low-priced stocks, and
- DJIA is calculated by a small sample of 30 stocks only which cannot be truly representative of the market as a whole.

Index calculation

b) Value-weighted indices – S&P 500:

The market values of all the companies included in the index are added together and divided by the market value of the companies on the day the index was first calculated multiplied by 100.

Suppose the S&P 500 has only two stocks, ABC and XYZ. Stock ABC has the initial price of $25 per share and the 20 million outstanding shares whereas XYZ has the initial price of $100 per share and the outstanding shares of 1 million. The sum of market value of stocks is_____.

If the price of ABC rises to $30 but that of XYZ drops to $90, find the sum of the new market value.

In this case, the value of the S&P 500 now is:

This method of calculating indices is better as high-priced stocks do not necessarily have a bigger impact on the index. It is market value that matters. Finally, as it is based on a sample of 500 stocks, the S&P 500 is more broad-based and thus more representative of the market as a whole.

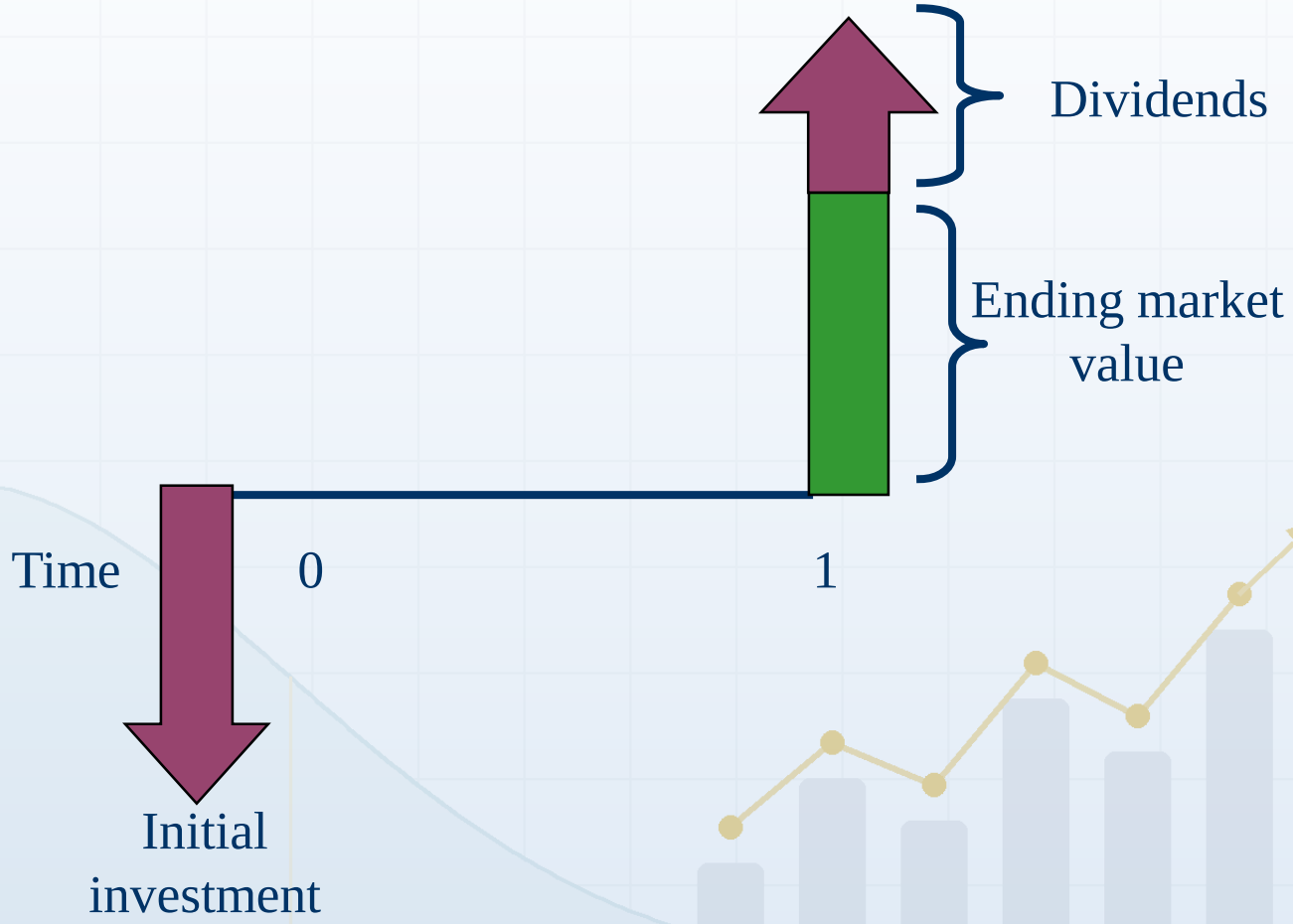
Classification of returns

- We will classify mean returns into **i) Historical realized returns and ii) Future (expected) returns.**
- **Historical returns** are simply the past performance of an asset such as the return of HSBC in 2020. We will first discuss how we can calculate I) the historical holding period returns, II) arithmetic and geometric average or mean returns and III) the measure of dispersion of historical returns.
- Future returns are returns of an asset in the future. Remember, on one knows the future return of an asset with certainty. We can at best estimate the average outcome returns known as expected (future) returns. Of course, historical returns may be used as a basis of the estimation of **expected returns**.

Historical returns of 1-period

1. **Dollar Returns**
   - the sum of the cash received and the change in value of the asset, in dollars.

2. **Percentage Returns**
   - the sum of the cash received and the change in value of the asset divided by the original investment.



Single period dollar returns

- **Total dollar returns**
  = income from investment such as dividend or interest payment
  + capital gain (loss) due to change in price of the investment

- Example:
  You bought a bond for $950 one year ago. You have received two coupons of $30 each. You can sell the bond for $975 today. What is your total dollar return?

- Interest payment = _____
- Capital gain = _____
- Total dollar return = _____

Single period percentage returns (Holding period returns)

How do we measure the total realized percentage return of a stock in a period?

$$R_{t+1} = \frac{D_{t+1} + P_{t+1} - P_t}{P_t} = \frac{D_{t+1}}{P_t} + \frac{P_{t+1} - P_t}{P_t}$$

= Dividend Income Yield + Capital Gain Yield

Example

You bought HSBC for $70 per share and you received dividends of $2.5 per share. The stock is sold for $75 per share in a year.

What is your total dollar return per share?

Dollar return = _____



What is your total percentage return?

Dividend yield = _____

Capital gain yield = _____

Total percentage return = _____

Average Annual Returns

The **average annual return** (%) of an investment during some historical period is simply the average of the realized percentage returns for each year.

There are two ways to calculate the average annual return.
I) Arithmetic mean return
II) Geometric mean return

Arithmetic Average (mean) Returns

- **Arithmetic average return or arithmetic mean (AM) return** is calculated by summing the sub period returns and dividing by the number of sub periods. Its formula is:

$$\bar{R} = \frac{1}{T}(R_1 + R_2 + \cdots + R_T) = \frac{1}{T}\sum_{t=1}^{T} R_t$$

Geometric Average (mean) Returns

- **Geometric average return or geometric mean (GM) return** is used to calculate the average rate per period on an investment that is compounded over multiple periods.

$$Geometric\ Return = \sqrt[n]{(1+r_1) \times (1+r_2) \times \ldots (1+r_n)} - 1$$

$r = rate\ of\ return$

$n = number\ of\ periods$

- The geometric average will be (more / less) than the arithmetic average unless all the returns are equal.

Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2005–2021 (Part 1)

Year End	S&P 500 Index	Dividends Paid*	S&P 500 Realized Return	Microsoft Realized Return	1-Month T Bill Return
2004	1211.92	-	-	-	-
2005	1248.29	23.15	4.9%	−0.9%	3.0%
2006	1418.30	27.16	15.8%	15.8%	4.8%
2007	1468.36	27.86	5.5%	20.8%	4.7%
2008	903.25	21.85	–37.0%	−44.4%	1.5%
2009	1115.10	27.18	26.5%	60.5%	0.1%
2010	1257.64	25.44	15.1%	−6.5%	0.1%
2011	1257.60	26.60	2.1%	−4.5%	0.0%
2012	1426.19	32.67	16.0%	5.8%	0.1%

Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2005–2021 (Continued)

* Total dividends paid by the 500 stocks in the portfolio, based on the number of shares of each stock in the index, adjusted until the end of the year, assuming they were reinvested when paid.

Source: Standard & Poor's, Microsoft, and U.S. Treasury Data

Year End	S&P 500 Index	Dividends Paid*	S&P 500 Realized Return	Microsoft Realized Return	1-Month T-Bill Return
2013	1848.36	39.75	32.4%	44.3%	0.0%
2014	2058.90	42.47	13.7%	27.6%	0.0%
2015	2043.94	43.45	1.4%	22.7%	0.0%
2016	2238.83	49.56	12.0%	15.1%	0.2%
2017	2673.61	53.99	21.8%	40.7%	0.8%
2018	2506.85	49.54	−4.4%	20.8%	1.7%
2019	3230.78	65.39	31.5%	57.6%	2.1%
2020	3756.07	69.14	18.4%	42.5%	0.4%
2021	4766.18	68.08	28.7%	52.5%	0.0%

Exercise

- Find the arithmetic mean return and the geometric mean return of S&P 500 Index from period 2005 to period 2021.

Historical risk of an investment

- **Risk** is the chance that the actual return from an investment may differ from what is expected. Usually we use **sample standard deviation** (also called **volatility**) of returns, a measure of dispersion of returns, on an asset to measure its risk.
- Historical sample variance of an asset's return

= (sum of squared deviations from the arithmetic mean)/(the number of observations – 1)

$$Var(R) = \frac{1}{T-1}\sum_{t=1}^{T}(R_t - \overline{R})^2$$

- Sample Standard deviation = square root of the sample variance

Exercise

- The mutual fund manager needs to estimate a security. Its performance during the last five years is shown in the table below.

	Year 1	Year 2	Year 3	Year 4	Year 5
Security A	2.3%	8.0%	-3.0%	5.5%	4.1%

- Find the sample S.D. of returns on Security A.

Volatility of U.S. Small Stocks, Large Stocks (S&P 500), Corporate Bonds, and Treasury Bills, 1926–2021

Investment	Return Volatility (Standard Deviation)
Small stocks	38.6%
S&P 500	19.7%
Corporate bonds	6.3%
Treasury bills	3.1%

Expected (future) returns and risk

- Since future returns are not known when we make investments, estimating return and risk of the investments are highly difficult.
- In finance, we usually assume there is a probability distribution governing the occurrence of the return.
- The probability distribution assigns a probability that each possible return will occur.
- How to calculate the expected return and variance / standard deviation of returns from a probability distribution?



Expected returns – single asset

Expected Return

$$E(R) = \Sigma p_i R_i$$

Where:

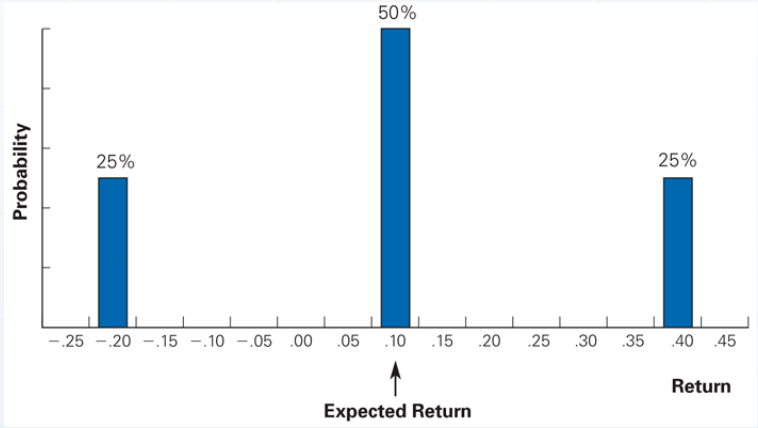
p_i = probability of the ith scenario

R_i = forecasted return in the ith scenario

Assume a stock currently trades for $100 per share.
In one year, there is a 25% chance the share price will be $140, a 50% chance it will be $110, and a 25% chance it will be $80.
Find the expected return on the stock.

Probability distribution of returns for a stock

Current Stock Price ($)	Stock Price in One Year ($)	Probability Distribution	
		Return, R	Probability, P_R
	140	0.40	25%
100	110	0.10	50%
	80	−0.20	25%



Expected return of the stock =

Variance and Standard deviation

Variance and standard deviation are used to measure the **risk** of a probability distribution.

Since the probability of each state is not necessarily the same, we can't simply divide the sum of squared deviations by the number of states. Instead, use the **probability as the weight** of each state.

In finance, the standard deviation of a return is also referred to as its **volatility.**

The standard deviation is easier to interpret because it is in the **same units as the returns themselves.**

Variance and Standard deviation

Variance: The expected squared deviation from the mean

$$\sigma^2 = \sum p_i[R_i - E(R)]^2$$

Standard Deviation, the square root of the variance, is

$$SD(R) = \sqrt{Var(R)}$$

Example – Standard deviation and Variance

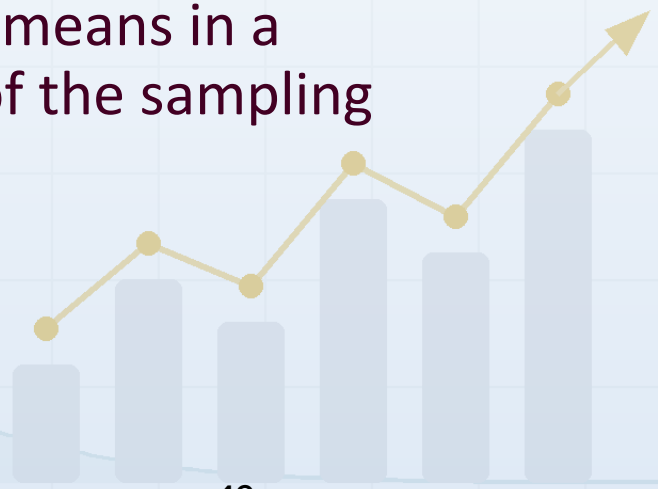
- Find the variance and standard deviation of returns on the stock on p 45.



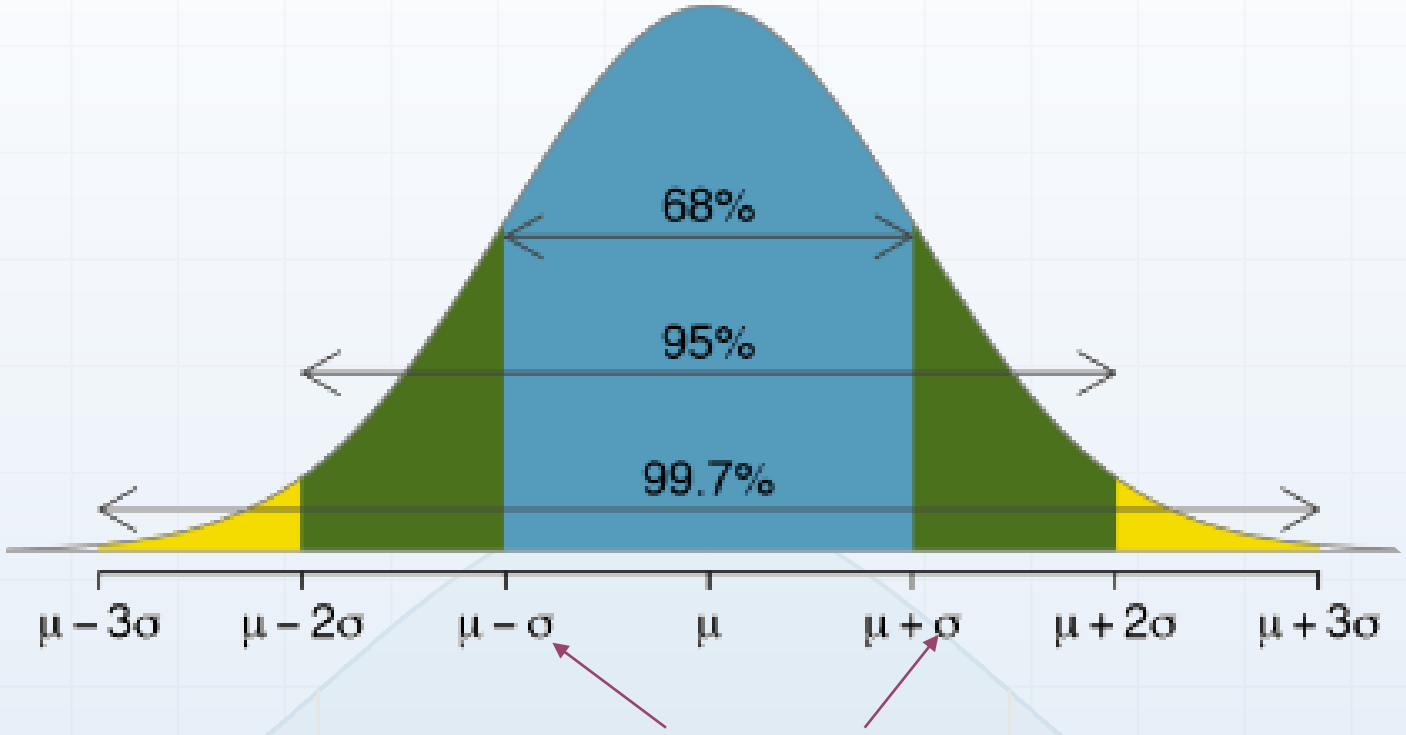
Estimation Error: Using Past Returns to Predict the Future

- We can use a security's historical average return to estimate its actual expected return. However, the average return is just an estimate of the expected returns. Seldom does the next year (future) return of an asset equal the historical mean return. Can we have an interval estimate of returns with a certain degree of confidence?

- The **standard error** is a measure of the variability of sample means in a sampling distribution. It represents the standard deviation of the sampling distribution of a statistic, such as the sample mean.



Distribution of the sample mean for a specified sample size



Note that the standard error (SE) of the mean estimate represents the standard deviation of the sampling distribution for the sample mean.

Estimation Error: Using Past Returns to Predict the Future

- If the distribution of a stock's return is identical each year, and each year's return is independent of prior years' returns, then the standard error of the expected return is:

$$Standard\ error\ of\ the\ estimate\ of\ the\ expected\ return = \frac{SD\ of\ data\ points}{\sqrt{\text{Number of Observations}}}$$

Estimation Error: Using Past Returns to Predict the Future

- A confidence interval for a sample mean provides a range of values within which the true population mean is likely to fall, with a certain level of confidence.

- The 95% confidence interval of the mean is:

$$P\left(\mu - 1.96 * \frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + 1.96 * \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

But if we rearrange this equation to solve for μ, we get…

$$P\left(\bar{X} - 1.96 * \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 * \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Estimation Error: Using Past Returns to Predict the Future

- The 95% confidence interval for the expected return can be re-written as

$$\text{Historical Average Return} \pm (1.96 \times \text{Standard Error})$$

- To simplify calculations, many individuals round 1.96 up to 2. The resulting **approximate 95% confidence interval** is:

$$\text{Historical Average Return} \pm (2 \times \text{Standard Error})$$

Example

- For the S&P 500 (1926–2021), there were 96 years of data. The average return of S&P 500 was 12.2% whereas its standard deviation of returns was 19.7% in the period. What is the approximate 95% confidence interval you would estimate for the S&P 500's expected return?
- Answer:

The approximate 95% confidence interval for the expected return is:

$$12.2\% \pm 2\left(\frac{19.7\%}{\sqrt{96}}\right) = 12.2\% \pm 4.0\%$$

Or a range from 8.2% to 16.2%

Conclusion

- Even though we have a long history of stock market, yet we cannot estimate the expected return (future return) of stocks or market very accurately.